

Jashore University of Science and Technology

Jashore-7408, Bangladesh



Guided Unsupervised Segmentation via Cross-Model Prompting

Submitted By:

Sabbir Ahmed

Student ID: 190110

Registration Year: 2019-2020

Supervisor:

Dr. A F M Shahab Uddin

Assistant Professor

Department of Computer Science and Engineering

This thesis report was submitted in partial fulfilment of the
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Faculty of Engineering and Technology

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Candidate's Declaration

This is to certify that the work presented in this thesis titled, “**Guided Unsupervised Segmentation via Cross-Model Prompting**” is the result of the research conducted by me under the supervision of Dr. A F M Shahab Uddin, Assistant Professor, Department of Computer Science and Engineering.

It is also announced that this thesis or any part of it has not been submitted anywhere else for the award of any degree, diploma, or other qualification.

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Sabbir Ahmed

Student ID: 190110

Session: Registration Year: 2019-2020

Department of Computer Science and Engineering

Jashore University of Science and Technology

Jashore-7408, Bangladesh

Certification

This is to certify that the thesis proposal entitled “**Guided Unsupervised Segmentation via Cross-Model Prompting**” has been submitted by Student ID **190110** in partial fulfillment of the requirements for the degree of B.Sc. Engineering in Computer Science and Engineering, under the Faculty of Engineering and Technology, Jashore University of Science and Technology, Jashore-7408, Bangladesh.

I would like to clarify that the author will complete their work under my supervision.

.....

Supervisor

Dr. A F M Shahab Uddin

Assistant Professor,

Department of Computer Science and Engineering

Jashore University of Science and Technology

Jashore-7408, Bangladesh

Dedication

This thesis is wholeheartedly dedicated to my family. Throughout the long and often challenging journey of this research, your unwavering support served as my constant anchor. Your profound belief in my abilities, even when my own confidence faltered, acted as a guiding light, illuminating the path forward. I am deeply aware of the countless sacrifices, both large and small, that you willingly made, and the immense patience you extended during demanding times. The steadfast foundation of love and understanding you provided created the space necessary for me to pursue and ultimately complete this work. This achievement, therefore, feels less like my own and more like a shared success, built upon your faith and generosity. For everything, I am eternally grateful.

Thesis Approval

This is to certify that the thesis work submitted by Sabbir Ahmed, entitled “**Guided Unsupervised Segmentation via Cross-Model Prompting**”, has been approved by the board of examiners of the requirements for the degree of B.Sc. in Engineering in the Department of Computer Science and Engineering, Jashore University of Science and Technology, Jashore, Bangladesh, in May 2025.

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Abstract

This thesis explores unsupervised medical image segmentation on a brain tumor dataset derived from the BraTS challenge, addressing the limitation of scarce annotated medical data. We investigate if large vision models pre-trained on natural images can provide useful segmentations without domain-specific fine-tuning. Initial experiments applied three pre-trained **OneFormer** models (COCO, ADE20K, Cityscapes) directly to 1500 selected BraTS image slices in a zero-shot manner. Performance was poor, highlighting a significant domain gap, as models failed to identify tumor regions accurately. However, a rare but critical observation was made: the COCO-trained OneFormer occasionally misclassified tumor areas as "sports ball," suggesting a weak shape-based correlation. We hypothesized that this erroneous mask could serve as a useful visual prompt. Testing this, we used specific "sports ball" masks generated by OneFormer to guide the **SegGPT** model on corresponding BraTS images. Quantitative analysis using Dice and IoU metrics showed variability based on the prompt used, but confirmed the generation of non-trivial segmentations. Qualitatively, this guided approach yielded substantially better segmentation of the tumor compared to the initial unsupervised OneFormer results. The findings demonstrate that even rare, domain-shifted misclassifications from general models can be repurposed as effective prompts for other architectures like SegGPT, offering a novel strategy to enhance unsupervised segmentation in specialized domains like medical imaging.

Chapter 1

Introduction

Chapter 1: Introduction

A key component of contemporary medical image analysis is medical image segmentation, which is the process of dividing digital medical images into several segments that each represent distinct objects or regions (such as organs, tissues, or diseases). Numerous therapeutic applications, such as computer-aided diagnosis, treatment planning (such as radiotherapy targeting), quantitative analysis of anatomical features, and tracking the course of a disease over time, depend on accurate segmentation. Brain tumor segmentation is a well-known instance where accurate tumor subregion delineation is essential for clinical decision-making and patient outcome assessment, especially when employing the multimodal MRI data frequently obtained via the BraTS challenge [30].

Although supervised deep learning techniques, particularly those derived from the U-Net architecture, have shown impressive results in attaining high segmentation accuracy on tests such as BraTS, their effectiveness is largely dependent on the availability of sizable datasets with accurate, voxel-level annotations supplied by qualified radiologists. Such annotated datasets require specific clinical expertise and are extremely costly, time-consuming, and labor-intensive to create. The development and implementation of supervised models are severely hampered by this **annotation bottleneck**, particularly for rare diseases or in environments with limited resources when sizable annotated cohorts are not available.

Therefore, there is a significant incentive to investigate different strategies that lessen reliance on large amounts of labeled data. By utilizing knowledge gathered from easily accessible large-scale datasets, frequently from many domains like natural pictures (e.g., COCO, ADE20K, Cityscapes), unsupervised and zero-shot learning paradigms present viable solutions. Without requiring any labels or annotations from the target medical domain, this thesis explores the viability of using strong, general-purpose vision models that have already been trained on such natural picture datasets to the job of medical image segmentation, notably brain tumors.

In our first investigation, we immediately used cutting-edge pre-trained segmentation models to unlabeled brain tumor photos. We tested variations pre-trained on COCO, ADE20K, and Cityscapes and chose OneFormer [22], a Transformer-based architecture with unified segmen-

tation capabilities. Unfortunately, these tests, which were carried out in a zero-shot fashion spanning instance, panoptic, and semantic tasks, produced subpar tumor segmentation findings. The models frequently produced segmentations that reflected the object categories they had acquired during their natural picture pre-training, failing to recognize or precisely outline tumor locations. This result clearly demonstrated the serious domain gap issue.

Nevertheless, an unexpected and uncommon trend surfaced during a thorough manual review of the many, mostly failed outputs produced by the COCO-pretrained OneFormer. A unique segmentation mask with the label **sports ball** was found to spatially overlap with the actual tumor location in a small number of cases (12 out of 1500 analyzed photos) (confirmed against ground truth). It implied that the visual aspects of the tumor were weakly activating some geometric features (possibly roundness or texture) that the model had learnt from natural images, despite the semantic absurdity and rarity of this happening.

The main hypothesis of this work was inspired by this coincidental observation: may this particular, semantically wrong but geographically relevant mask function as a helpful, if feeble, signal or visual suggestion for a different kind of segmentation model? We changed course and used SegGPT [39], a paradigm for prompt-based, in-context segmentation, to look into this. By providing SegGPT with the target brain tumor image and the extracted "sports ball" mask from OneFormer as a visual cue, we effectively instructed SegGPT to identify areas in the target image that resembled this cue. The outcomes were striking: SegGPT generated qualitatively better tumor structure segmentations based on this weak data than the original unsupervised OneFormer outputs. The methodology, qualitative and quantitative results of this two-phase study, and possible ramifications of using such unusual cross-domain signals to advance unsupervised medical picture segmentation are all included in this thesis.

1.1 Background

The goal of image segmentation, a basic computer vision problem, is to divide a picture into discrete, meaningful areas or objects. In the medical field, this means recognizing diseased regions (like lesions or tumors) or anatomical features (like organs or veins) in medical scans like Computed Tomography (CT) or Magnetic Resonance Imaging (MRI). Precise medical picture segmentation is essential because it yields quantitative data that aids in clinical diagnosis, permits accurate treatment planning (such as radiation therapy or surgical navigation), and permits objective tracking of the disease's reaction to therapy. The most popular method for supervised medical segmentation is deep learning models, especially Convolutional Neural Networks (CNNs) based on the U-Net architecture and its several variations. These models frequently achieve state-of-the-art performance on difficult benchmarks like the BraTS dataset [30] for segmenting brain tumors. The great performance of these supervised techniques, however, depends on extensive datasets with painstaking, voxel-level annotations, which continue to be a major bottleneck because of the expense and level of experience needed.

Research into less supervised approaches is encouraged by this limitation. With zero-shot learning, models learned on one domain (like natural photos) are applied directly to another (like medical images) without target-domain examples, whereas unsupervised learning seeks to identify patterns without labels. The significant disparities in statistics and semantics between the source and target domains, known as the "domain gap," are a significant obstacle in zero-shot transfer. Unifying semantic, instance, and panoptic segmentation, models such as **OneFormer** [22] have strong visual representation capabilities and have been pre-trained on large, heterogeneous natural image datasets (COCO, ADE20K, Cityscapes). But when these models are applied directly to specific medical images, they frequently fail because the learned features might not match the pertinent medical structures. Parallel to this, models such as **SegGPT** [39] reflect prompt-based or in-context segmentation and are built on foundational structures [38]. These models, which are comparable to models like Segment Anything [24] but frequently con-

centrate on contextual transfer, offer flexibility beyond fixed semantic categories by learning to segment based on visual similarity to a prompt picture or mask. This thesis explores how these ideas intersect: can a generalist model like OneFormer's failure mode on medical data serve as a helpful suggestion for a model like SegGPT, bridging the domain gap in an unsupervised way.

1.2 Problem Statement

The development of high-performance medical image segmentation models using supervised deep learning is severely hampered by the high cost, time, and expert knowledge required to create large, accurately annotated datasets. This annotation bottleneck limits the scalability and applicability of supervised methods, particularly for diverse pathologies or in low-resource clinical environments. While leveraging powerful vision models pre-trained on abundant natural image datasets presents an attractive alternative, their direct application to medical images in a zero-shot setting typically fails. This failure stems from the significant domain shift between natural images (containing everyday objects, complex scenes, diverse textures) and medical images (often grayscale or specific pseudo-color maps, featuring unique anatomical structures, subtle pathological textures, and distinct noise characteristics). Features learned to recognize cars, buildings, or animals often do not readily map to identifying tumors, lesions, or specific tissue types within the human body.

Existing solutions like unsupervised domain adaptation (UDA) often require access to **unlabeled target data** during an adaptation phase or rely on aligning global feature distributions, which may not capture fine-grained details. This thesis addresses the specific problem of how to effectively utilize pre-trained, general-purpose segmentation models for a specific target medical task (brain tumor segmentation using data derived from the BraTS challenge) without relying on target domain annotations or conventional domain adaptation training procedures. We explore the underexplored possibility that specific failure modes or misclassifications generated by these generalist models when confronted with out-of-domain medical data might, counterintuitively, contain latent, useful information (a "weak signal"). The core research problem is to

investigate if such weak signals can be identified and effectively repurposed as visual prompts to guide a different, context-aware segmentation model (SegGPT) towards producing meaningful and improved segmentations on the target medical domain.

1.3 Objectives

The primary goal of this undergraduate thesis is to investigate a novel approach for unsupervised medical image segmentation by leveraging cross-domain model interactions. The specific objectives are fourfold. To systematically evaluate the baseline zero-shot segmentation performance of multiple state-of-the-art OneFormer models, pre-trained on diverse natural image datasets (COCO, ADE20K, Cityscapes), when directly applied to the BraTS brain tumor dataset, assessing their inherent capability (or lack thereof) to segment tumors without domain-specific knowledge. Then, perform a detailed qualitative analysis of the failure modes exhibited by these models, specifically searching for and documenting any recurring or potentially informative misclassifications, such as the observed "sports ball" label coinciding with tumor regions. Based on these observations, to formulate and experimentally test the central hypothesis: whether a specific, semantically incorrect mask generated by OneFormer can function as an effective visual prompt to guide the SegGPT model towards segmenting the corresponding structure (the tumor) in the target BraTS image. Finally, to evaluate the outcomes of this guided SegGPT approach through both qualitative visual comparison against the baseline OneFormer results and quantitative assessment using standard segmentation metrics (Dice, IoU) against the ground truth, thereby determining the extent of improvement achieved by this weak-signal prompting strategy.

1.4 Research Questions

This study seeks to answer several key questions regarding the potential of unsupervised cross-domain segmentation using pre-trained models. Primarily, how effective are powerful, general-

purpose segmentation models like OneFormer, pre-trained on natural images, when applied directly in a zero-shot fashion to a vastly different domain like the BraTS medical imaging dataset? Secondly, beyond overall failure, can specific patterns within the incorrect outputs or misclassifications generated by such models (e.g., the anomalous "sports ball" label appearing on tumors) contain latent visual information or cues that are relevant, despite semantic inaccuracy, to the actual underlying structure in the target medical task? Thirdly, can a visually prompted segmentation model, such as SegGPT, successfully interpret and leverage such a potentially flawed or noisy mask, generated by a different model like OneFormer, as an effective visual guide to perform segmentation in the target domain? Finally, does utilizing this specific form of weak, cross-domain signal as a prompt lead to a demonstrable improvement in segmentation quality, assessed both qualitatively (compared to the baseline model's output) and quantitatively (in terms of overlap with ground truth using metrics like Dice/IoU), for the target structure (brain tumors).

1.5 Significance of the Study

This study may be important since it investigates a new, unorthodox approach to the enduring problem of data scarcity in medical image segmentation. This study provides a proof-of-concept for a highly resource-efficient method by showing that even outputs that appear to be useless—more specifically, infrequent misclassifications resulting from domain shift—generated by large pre-trained models can be recognized and used as helpful guidance signals for other models. If the underlying idea is generalizable, it offers a mechanism to harness the potential of foundation models that have been trained on publicly accessible data without requiring costly target-domain annotation or intricate domain adaption training processes. This might have a special effect on the use of cutting-edge AI methods in healthcare settings with limited resources.

Chapter 2

Literature Review

Chapter 2: Literature Review

Computer-aided diagnosis, treatment planning, and illness progression tracking all depend on medical image segmentation, which is the act of dividing medical images such as MRI or CT scans into discrete sections that correspond to various tissues, organs, or pathologies [27]. Brain tumor segmentation is a clinically important yet very difficult problem, especially when considering the multimodal Brain Tumor Segmentation (BraTS) challenge dataset [30, 2]. The segmentation of active tumor core, peritumoral edema, and necrotic/non-enhancing tumor core is necessary for the BraTS dataset, which usually consists of T1-weighted, T1-contrast-enhanced, T2-weighted, and FLAIR MRI scans. High variation in tumor shape, size, location, and appearance, heterogeneity among tumor subregions, unclear boundaries, and the complexity of surrounding brain structures are the main causes of the inherent difficulties.

Medical picture segmentation has been transformed by the introduction of deep learning, particularly Convolutional Neural Networks (CNNs). With its symmetric encoder-decoder structure and skip connections, the U-Net architecture [34] became the de facto standard. It gave rise to a number of variations, including V-Net [31] for 3D data, UNet++ [?] with nested and dense skip pathways, and Attention U-Net [32] with attention mechanisms. On benchmarks like BraTS, these models frequently perform on par with or better than human experts when trained on sizable datasets with pixel-level annotations [21]. However, the availability of large amounts of high-quality labeled data is a major prerequisite for this achievement. Such datasets are notoriously hard to create for medical imaging, requiring clinical professionals to devote a lot of time. This makes supervised techniques expensive and limits their application in certain situations with rare diseases, unusual pathologies, or a lack of readily available annotations. This lack of evidence encourages the investigation of different paradigms.

By creating pretext tasks like contrastive learning [5, 16], masked image modeling [15], or image rotation prediction, Self-Supervised Learning (SSL) has become a potent method for learning representations from unlabeled data. Large unlabeled medical datasets have been used to pre-train models in medical imaging using SSL. These models can subsequently be refined

on smaller labeled sets for certain tasks [45]. SSL nevertheless usually needs to be adjusted for the particular downstream segmentation requirement, even though it shows promise for utilizing unlabeled medical data.

The goal of Unsupervised Domain Adaptation (UDA) approaches is to bridge the gap between an unlabeled target domain (like clinical scans) and a label-rich source domain (like natural images, simulated data, or even data from a different scanner/protocol) [40]. Typical UDA techniques include minimizing statistical discrepancies such as Maximum Mean Discrepancy (MMD), employing self-ensembling and pseudo-labeling on the target domain, or aligning feature distributions through adversarial training (e.g., inspired by CycleGAN [46] for image-to-image translation or domain discriminators [12]). UDA frequently still needs access to a sizable amount of unlabeled target data during the adaptation training phase, despite its effectiveness in many situations.

With the help of training data and possibly supplementary information such as textual descriptions or semantic attributes, Zero-Shot Learning (ZSL) seeks to perform tasks on unseen classes or domains without the use of specific examples from those classes or domains [25, 41]. The significant domain gap—differences in image statistics, textures, structures, and pertinent item categories—makes it especially difficult to apply models trained only on natural images to medical segmentation in a zero-shot fashion. For complicated medical structures like tumor sub-regions, existing ZSL segmentation frequently depends on semantic embeddings or textual descriptions of target classes, which may not be easily accessible or adequately informative.

ADE20K [43], COCO [26], Cityscapes [9], ImageNet [10], and Vision Transformers (ViTs) [11] (building on the original Transformer [38]) are just a few of the large foundation models that have gained popularity in recent years. The state-of-the-art in unified segmentation (semantic, instance, panoptic) on these benchmarks is represented by models such as OneFormer [22], which show remarkable abilities in comprehending complicated scenes. This thesis' cen-

tral question is whether, in spite of the domain change, medical tasks can benefit from the rich visual representations that these models learn. Our preliminary trials and the literature examining the direct transfer of natural image models [6, 1] both support the first attempts at direct application, frequently demonstrate how these models fail to successfully identify clinically relevant structures like tumors in medical photos, instead choosing to recognize patterns that are pertinent to their underlying domain. Through advanced fine-tuning techniques or ongoing pre-training on sizable medical datasets, some works try to adapt such foundation models to the medical domain [14, 44]. However, this again departs from a truly zero-shot or annotation-free scenario for the particular target task.

At the same time, a new class of models has surfaced that allow for visually guided or prompt-based segmentation. For segmenting arbitrary objects based on point, box, or text prompts, the Segment Anything Model (SAM) [24] showed impressive zero-shot generalization. Following SAM, architectures such as SegGPT [39] further showed that they could effectively execute "in-context learning" for segmentation by segmenting a target image based on visual cues (exemplars) presented in a prompt image or mask. By depending on visual similarities rather than predetermined semantic labels for the objective task, this paradigm provides a flexible means of guiding segmentation. Though they also point out difficulties in fully automated or zero-shot environments, early studies on SAM's suitability for medical imaging have showed promise for some tasks, especially when interactive prompting is practical, for intricate organizations [29]. An option is provided by SegGPT's visual prompting mechanism, in which the guidance itself may originate from a variety of sources, such as possibly artificial exemplars or, as suggested here, outputs from other models.

Although the promise of foundation models and prompt-based learning, as well as the difficulties of domain adaptation, are well established, the precise tactic of using generalist model misclassifications as instructive cues for a target domain job is still mainly unknown. The majority of research focuses on enhancing model generalization and accuracy, frequently treating

inaccurate predictions as data points for out-of-distribution detection or as mistakes to be minimized [19]. Although they usually don't explicitly aim to repurpose the mistake signals as useful input for another model, certain efforts on model interpretability or explainability [35] try to understand why models make particular predictions, including errors. An innovative approach is the notion that a particular, albeit uncommon and semantically incorrect, output (such as a "sports ball" mask that coincides with a tumor) may encode valuable latent features (such as shape, boundary information, or salient region cues) that can be constructively repurposed as a prompt for another model. In contrast to approaches that might employ model uncertainty estimates to direct selective annotation or active learning, the substance of the misclassification—rather than merely its uncertainty—is thought to be relevant in this case.

By employing the OneFormer-to-SegGPT pipeline on a dataset acquired from BraTS, this thesis seeks to explore this area and assess if such "weak signals" emerging from the failure modes of cross-domain application can offer useful assistance for unsupervised medical picture segmentation. It explores a potentially resource-efficient way to adapt powerful vision models to specialized domains without the need for target-domain annotations or explicit domain adaptation training. It does this by combining prompt-based segmentation, zero-shot learning (since no BraTS labels are used for OneFormer training or SegGPT direct supervision), and the analysis of model behavior under domain shift.

Chapter 3

Methodology

Chapter 3: Methodology

The potential of using pre-trained vision models for unsupervised brain tumor segmentation was examined in this study using a two-phase experimental design. The main idea was to use a publicly available dataset from the BraTS challenge to evaluate the zero-shot capabilities of a state-of-the-art segmentation model (OneFormer) and then investigate whether a particular failure mode could be used to direct a different model (SegGPT) towards better performance.

3.0.1 Dataset and Preprocessing

The "Brain Tumor Image DataSet: Semantic Segmentation," which is openly accessible on Kaggle [30], serves as the main dataset used in this investigation. This dataset comes from the larger BraTS (Brain Tumor Segmentation) challenge project [30], which offers multimodal MRI scans of patients with brain tumors. It has been selected for semantic segmentation tasks. On August 19, 2023, the Roboflow platform exported the Kaggle dataset that was used in this study. Its description states that it includes 2146 photos with COCO segmentation format tumor labels. Class 0 (non-tumor) and Class 1 (tumor) are the two classes defined by the dataset. Some preprocessing processes were used as part of the Roboflow export process, such as extending photos to 640x640 pixels and automatically aligning pixel data. During exporting, no image enhancement was used.

From the 'train', 'test', and 'validation' splits offered inside the Kaggle dataset structure, a subset of 1500 image slices was chosen at random for this thesis. Processing the ground truth (GT) annotations was an important first step in our pipeline. The segmentation information for the 'Tumor' class was extracted by parsing the COCO-format JSON files that accompanied the photos. After that, this data was transformed into binary masks, with the tumor-corresponding pixels given a value of 1 and the background/non-tumor pixels given a value of 0. The structure of the original picture directory was replicated when these created binary GT masks were stored as distinct image files (such as PNG format) in a special 'mask' directory. To enable easy access during tests, the original picture filenames and the associated generated mask filenames were both carefully renamed to a simpler format (e.g., sequential numbering or using the original

base name). Figure 3.1 displays some image slices and the binary ground truth masks that were created from the dataset.

It is critical to emphasize that during the model inference phases (both Phase 1 with OneFormer and Phase 2 with SegGPT), only the raw image slices (after the initial Roboflow resize) were provided as input to the models. The generated ground truth binary masks residing in the ‘mask’ folder were strictly reserved for post-hoc analysis. They were used solely for visually verifying the location of the "sports ball" misclassifications from OneFormer and for the final quantitative evaluation (calculating Dice and IoU scores) of the SegGPT-generated segmentation masks against the actual tumor regions. No ground truth information was used to guide or train the models at any stage.

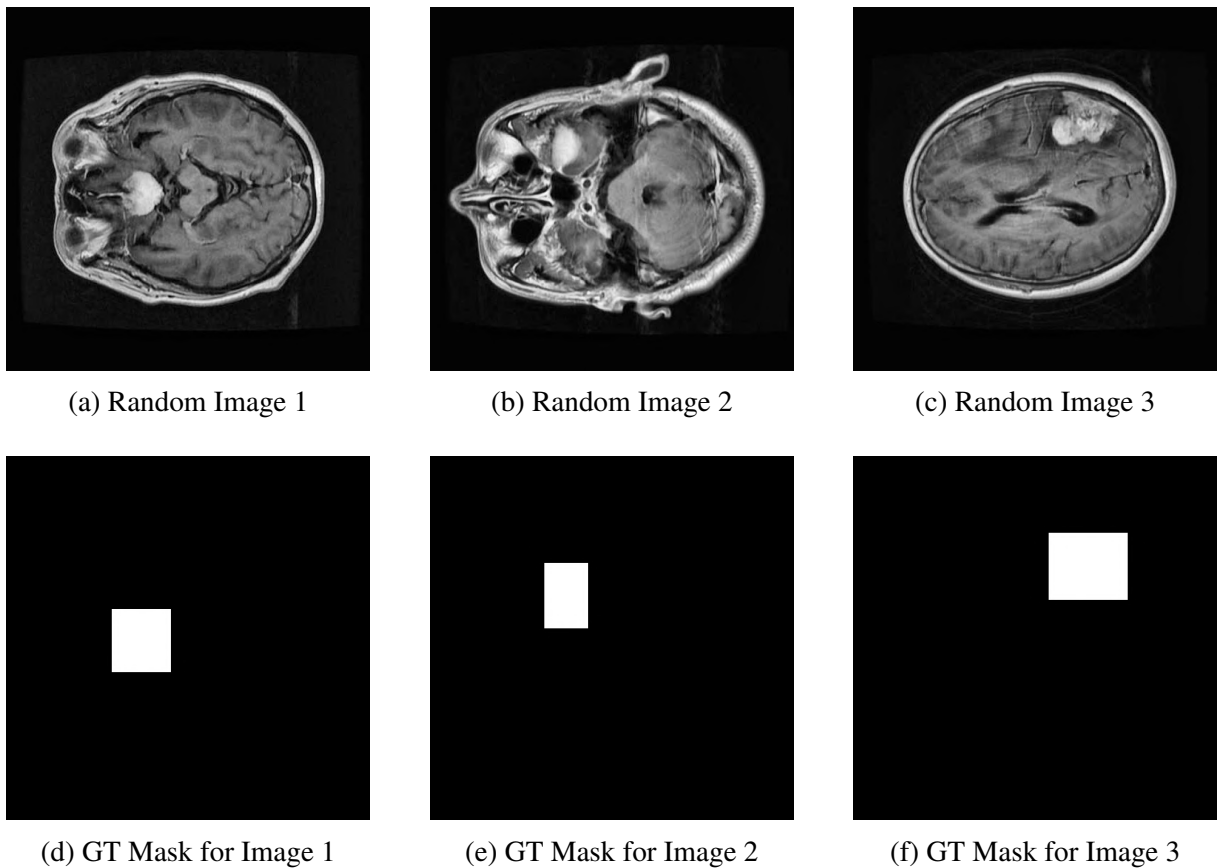


Figure 3.1: Examples of data used from the processed dataset [30]. Top row: Sample MRI image slices that has been selected randomly. Bottom row: Corresponding binary ground truth masks generated during preprocessing, where white indicates the tumor region (Class 1) and black indicates the background (Class 0). These masks were used only for evaluation.

3.0.2 Models and Experimental Setup

Two primary pre-trained models were utilized:

- **OneFormer:** We used pre-trained checkpoints for OneFormer [22] that are publically available, specifically three versions that were trained on different large-scale natural picture datasets: Cityscapes, ADE20K, and COCO. The corresponding codebase was used to load these models, which were acquired as ‘.pth’ files. With OneFormer, panoptic, instance, and semantic segmentation are all combined into a single Transformer-based design.
- **SegGPT:** A pre-trained version of the SegGPT model [39], designed for visually prompted segmentation, was employed in the second phase. SegGPT operates on the principle of "segment everything in context," meaning it takes a reference image (or mask) and a target image, and segments regions in the target that are visually similar to the reference. An example of its general operation on natural images is shown in Figure 3.2.

The software environment was managed using Conda. All experiments were conducted using Python as the primary programming language and PyTorch as the core deep learning framework. The Hugging Face Transformers library was potentially utilized for model loading or related functionalities associated with the pre-trained models. Visualization of results and intermediate analysis relied on libraries such as Matplotlib and Seaborn.

3.0.3 Phase 1: Zero-Shot Evaluation of OneFormer

The initial phase focused on evaluating the direct, unsupervised application of OneFormer to the selected 1500 BraTS-derived image subset. For each image slice, the raw image data was passed as input to each of the three pre-trained OneFormer models (COCO, ADE20K, Cityscapes). Inference was run to generate outputs for all available segmentation tasks supported by OneFormer: semantic segmentation, instance segmentation, and panoptic segmentation. This process treated the task as purely zero-shot, leveraging only the models’ pre-existing knowledge

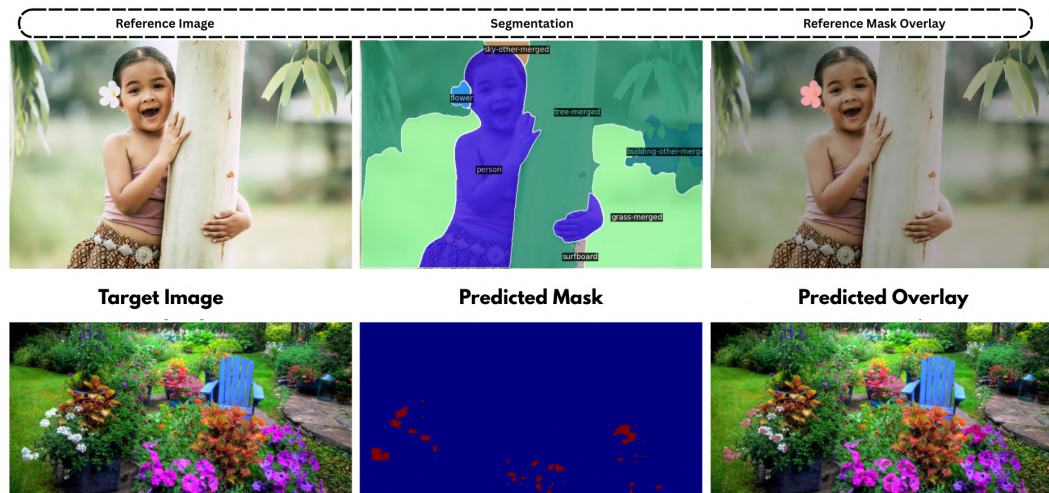


Figure 3.2: Illustration of SegGPT’s general visual prompting mechanism on natural images [39]. Given a reference image/mask pair (top row showing mask overlay), SegGPT segments corresponding objects or regions in a target image (bottom row showing prediction overlay). This demonstrates its ability to segment based on visual similarity rather than fixed semantic classes.

from natural images. The resulting segmentation masks and associated labels (reflecting categories from COCO, ADE20K, or Cityscapes) were saved. Initial analysis involved reviewing the large volume of these generated outputs across all models and tasks.

3.0.4 Phase 2: Observation, Hypothesis, and SegGPT Experiment

During the manual review of the outputs from Phase 1, with a focus on the semantic segmentation results from the COCO-pretrained OneFormer, the rare but significant "sports ball" misclassification was observed. Approximately 12 instances of this mask were found across the 1500 images processed, and critically, these "sports ball" masks were visually confirmed to spatially overlap with the known tumor regions by comparing their location to the ground truth masks derived from the dataset annotations [30].

This observation prompted the hypothesis that these specific, semantically incorrect masks might contain useful visual cues recognizable by a visually-prompted model. To test this, the

second experimental phase used SegGPT. For a set of target BraTS-derived images, the following procedure was executed using each of the three selected "sports ball" prompt masks:

1. The raw target image slice served as the main input to SegGPT.
2. One of the three selected binary masks ("sports ball" prediction from '2829.jpg', '281.jpg', or '1112.jpg') was provided as the visual reference prompt.
3. SegGPT was run in its visually prompted inference mode.

This process was repeated for multiple target images using each of the three distinct prompts to assess the consistency and quality of the resulting segmentations.

3.0.5 Evaluation Approach

The evaluation incorporated both qualitative and quantitative measures. Qualitatively, the segmentation masks generated by SegGPT in Phase 2 were visually compared against the baseline OneFormer outputs for the same tumor regions, assessing subjective improvements in boundary delineation and relevance. Quantitatively, for the segmentations generated by SegGPT using the three specific prompts ('2829.jpg', '281.jpg', '1112.jpg'), the Intersection over Union (IoU) and Dice Similarity Coefficient (Dice) scores were calculated by comparing the generated mask corresponding to the tumor region against the ground truth binary mask derived during preprocessing from the dataset annotations [30]. The distributions of these scores, along with their mean and median values, were analyzed using histograms and box plots to provide a quantitative measure of overlap accuracy for each prompting condition.

3.1 Research Design

The research employed a sequential, two-stage experimental design. Stage one established a zero-shot baseline using multiple pre-trained OneFormer models on the selected BraTS-derived dataset. Stage two tested a specific hypothesis derived from observations in stage one, using a

visually prompted model (SegGPT) guided by particular outputs from OneFormer. Evaluation involved both qualitative comparison and quantitative metrics (IoU, Dice) against ground truth for the SegGPT outputs.

3.2 Data Collection

The data for this study was sourced from the "Brain Tumor Image Data Set : Semantic Segmentation" available on Kaggle [30]. This dataset, derived from the BraTS challenge [30] and exported via Roboflow, consists of 2146 MRI image slices with tumor annotations in COCO format. For our experiments, a working subset of 1500 image slices was randomly selected. The primary data collection step involved processing the provided COCO annotations to generate binary ground truth (GT) masks for the tumor class (Class 1 vs. Class 0). These GT masks, along with the original images (resized to 640x640 during Roboflow export), were systematically organized and renamed. As detailed previously (Section 3.1), the GT masks were explicitly reserved for post-hoc evaluation and were not used during any model inference stage.

3.3 Implementation

The experiments were implemented using Python within a Conda environment for managing dependencies. The core deep learning operations relied on the PyTorch library. The official codebases for OneFormer [22] and SegGPT [39] were adapted, potentially utilizing components from the Hugging Face Transformers library for model handling. Pre-trained model weights (‘.pth’ files) for OneFormer (COCO, ADE20K, Cityscapes versions) and SegGPT were used as provided by their respective authors. Data manipulation and result visualization were aided by libraries such as Matplotlib and Seaborn. Additional libraries used included Detectron2, NumPy and OpenCV (refer to Table 3.1 for specific versions). All computations were performed in the VMI Lab on a workstation equipped with dual NVIDIA GeForce RTX 2070 Super GPUs and 32 GB of system RAM. Generated segmentation masks and qualitative results were saved and analyzed manually and programmatically.

Library	Version
PyTorch	2.5.1
Detectron2	0.6
Matplotlib	3.10.1
Seaborn	0.13.2
Transformers	4.51.3
NumPy	2.2.4
OpenCV (cv2)	4.11.0

Table 3.1: Key Python Libraries and Versions Used

Chapter 4

Results and Discussion

Chapter 4: Results and Discussion

This chapter presents the outcomes of the experiments detailed in the Methodology section. It first describes the results obtained from the initial zero-shot application of OneFormer models, followed by the qualitative and quantitative findings from the guided SegGPT experiment prompted by specific "sports ball" masks. Subsequently, it discusses the interpretation of these results and their implications.

4.1 Results

The experimental investigation yielded distinct results in its two phases, incorporating both qualitative observations and quantitative metrics for the guided segmentation phase.

Phase 1: OneFormer Zero-Shot Performance. The direct application of the three pre-trained OneFormer models (COCO, ADE20K, Cityscapes) to the 1500 selected BraTS-derived image slices [30] produced segmentation outputs that were qualitatively judged as poor for the task of identifying brain tumors. The models generated numerous segmentation masks, but these did not accurately delineate tumor regions and were labeled with irrelevant natural-image categories, confirming the challenge of zero-shot domain transfer. Figure 4.1 illustrates typical examples of the varied but generally unhelpful segmentations produced by the COCO-OneFormer across different image slices.

During manual inspection of the 23,000 masks generated by the COCO-OneFormer, approximately 12 instances were identified where a mask labeled "sports ball" spatially coincided with the ground truth tumor location (verified against GT masks derived from [30]). Figure 4.2 shows a specific example where the "sports ball" mask overlaps the tumor area, alongside other irrelevant segmentations. This rare occurrence became the focal point for the next phase.

To further verify the spatial correlation of this specific misclassification, Figure 4.3 compares the original image, the ground truth tumor annotation overlay, and the OneFormer-predicted "sports ball" mask overlay for several instances where this occurred. This visual confirmation validates that the generated "sports ball" masks consistently localized within the tumor region across these rare cases.

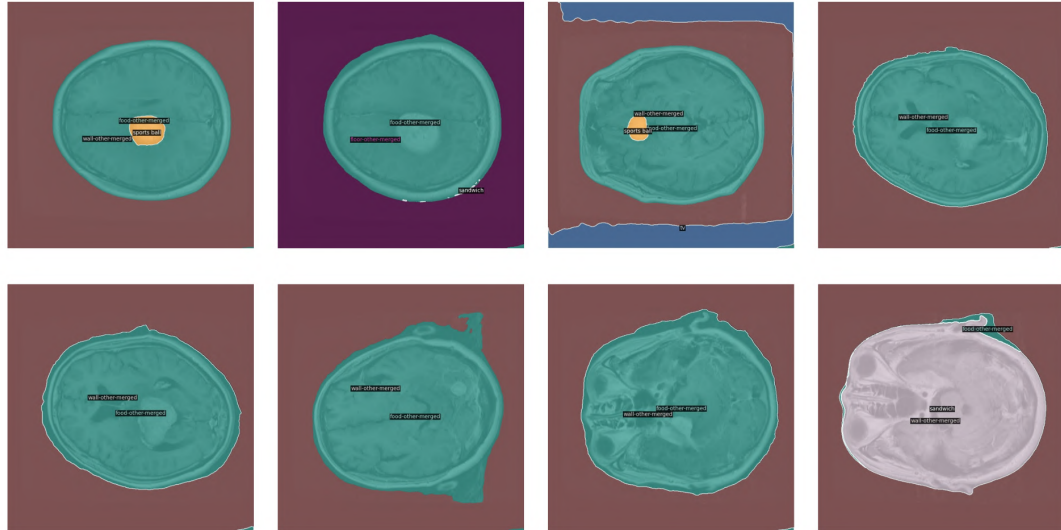


Figure 4.1: Failed OneFormer Zero Shot Prediction examples. Semantic segmentation results from COCO-pretrained OneFormer on various image slices from the dataset [30]. The generated masks (e.g., 'wall-other-merged', 'food-other-merged') reflect natural image categories and fail to accurately segment the tumor regions, demonstrating the domain gap.

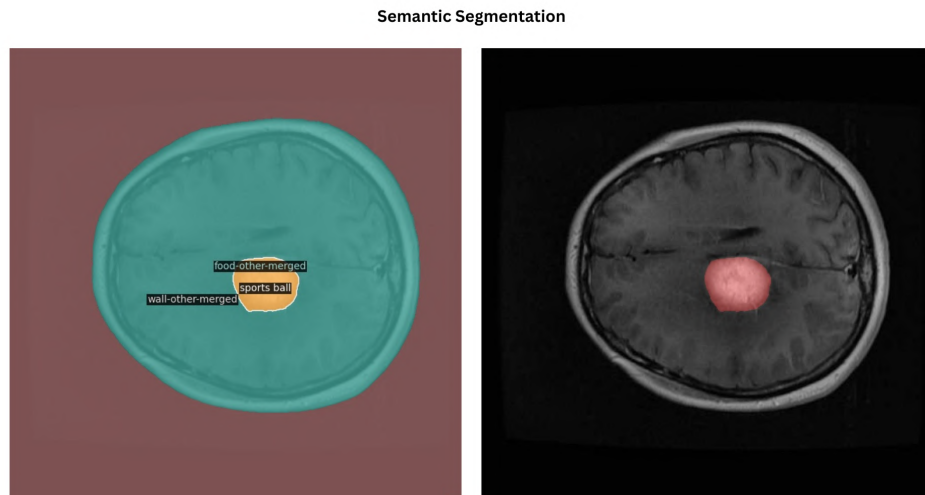


Figure 4.2: Example of COCO-OneFormer semantic segmentation output on an image slice. The erroneous "sports ball" label (e.g., red mask (right picture)) coinciding with the tumor location, amidst other irrelevant labels. This highlights the specific misclassification used for prompting.

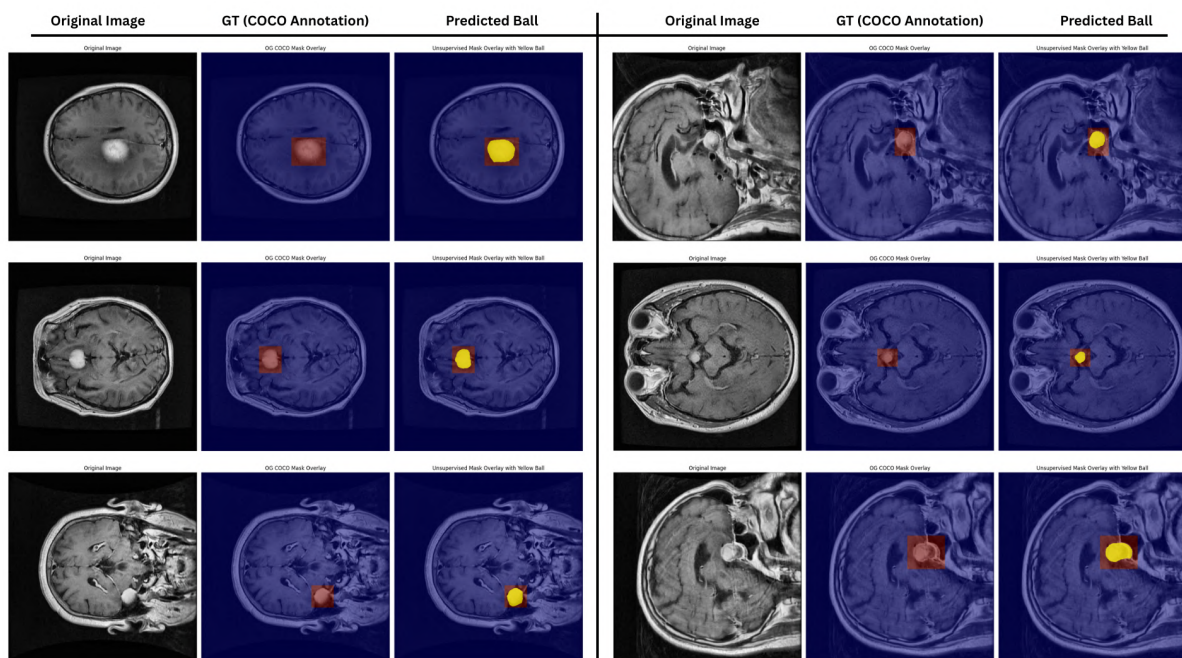


Figure 4.3: Visual verification of the "sports ball" misclassification. Each row shows: Original image slice (left), Ground Truth tumor overlay (center, e.g., red), and the corresponding "sports ball" mask predicted by COCO-OneFormer (right, e.g., yellow). These examples demonstrate the consistent spatial overlap between the erroneous prediction and the actual tumor across different cases.

Phase 2: Guided SegGPT Performance. This phase investigated the effectiveness of using these specific "sports ball" masks as visual prompts for SegGPT. Three representative prompt masks, derived from OneFormer’s output on images ‘2829.jpg’, ‘281.jpg’, and ‘1112.jpg’ from the dataset [30], were used to guide SegGPT segmentation on multiple target image slices.

Qualitative Results: Across the cases tested, the segmentations produced by SegGPT when guided by the "sports ball" prompts were consistently judged qualitatively superior to the baseline zero-shot OneFormer outputs for the corresponding tumor regions. The SegGPT masks generally captured the tumor shape and boundaries more accurately, appearing more coherent and clinically relevant. Figure 4.4 showcases examples of this improvement, illustrating how the prompt mask guides SegGPT to produce a targeted tumor segmentation (green overlay) that aligns better with the ground truth (red overlay) compared to the initial OneFormer attempt.

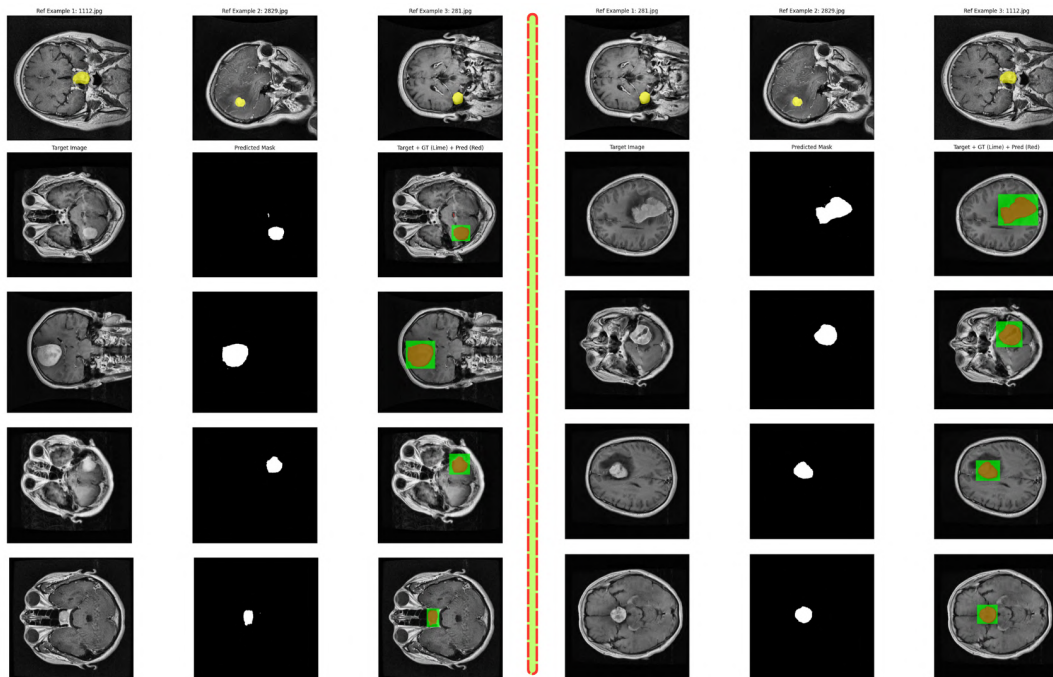


Figure 4.4: Qualitative results of the guided SegGPT experiment. Each row represents a different target image slice. Columns show: Reference prompt image (often the source of the 'sports ball' mask), Target image slice, Binary mask predicted by SegGPT, and Overlay of Target + Ground Truth (Green Box) + SegGPT Prediction (Red blob). These examples illustrate SegGPT generating tumor segmentations (red) guided by the prompt, achieving better localization than baseline OneFormer.

Quantitative Results: To quantify the accuracy of the SegGPT outputs relative to ground truth, IoU and Dice scores were computed for segmentations generated using each of the three specific prompts. A notable variability in performance was observed depending on the specific prompt used. Figures 4.5 and 4.6 visually present the source images and the corresponding "sports ball" masks that served as the best and worst performing prompts, respectively. These visualisations help contextualize the inputs that led to the divergent quantitative outcomes discussed below, potentially hinting at subtle differences in the prompt mask quality or the complexity of the source image scene.

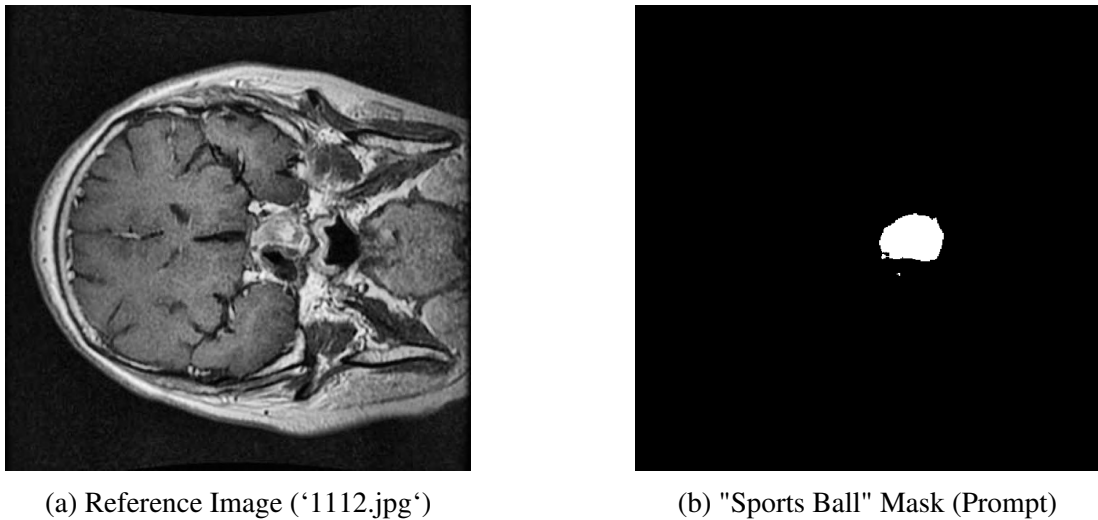
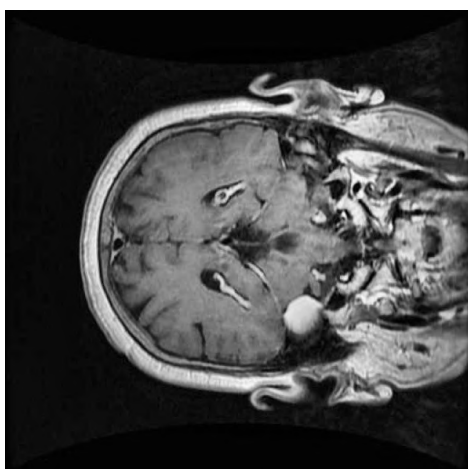


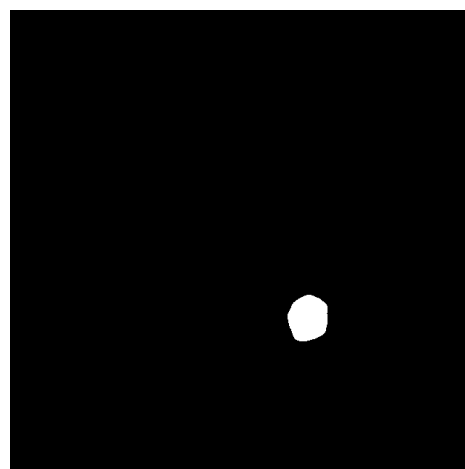
Figure 4.5: Source of the Best Performing Prompt. Left: The original image slice ('1112.jpg') from which the prompt was derived. Right: The binary "sports ball" mask generated by COCO-OneFormer for this image. This mask, when used as the visual prompt for SegGPT, led to the highest average quantitative scores (Dice/IoU) compared to other prompts tested.

We focus here on comparing the best and worst performing prompts based on average scores. The prompt derived from image '**1112.jpg**' (source shown in Figure 4.5) yielded the best quantitative results, achieving a mean Dice score of 0.316 (median 0.253). In contrast, the prompt derived from image '**281.jpg**' (source shown in Figure 4.6) performed the worst quantitatively, with a mean Dice score of 0.120 (median 0.000). This difference is clearly visualized by comparing the score distributions. Figure 4.7 presents the Dice score histograms side-by-side for the best ('1112.jpg') and worst ('281.jpg') performing prompts.

The box plots provide another view of this variability. Figure 4.8 compares the distribution

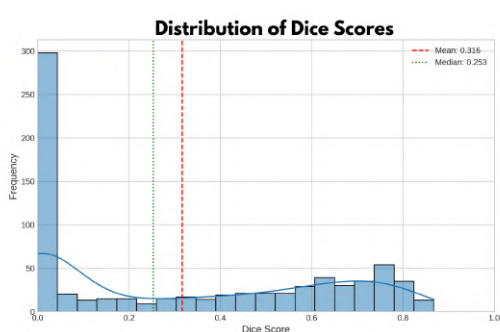


(a) Reference Image ('281.jpg')



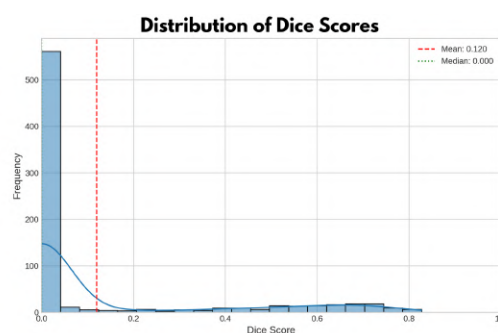
(b) "Sports Ball" Mask (Prompt)

Figure 4.6: Source of the Worst Performing Prompt. Left: The original image slice ('281.jpg') that generated the prompt. Right: The corresponding binary "sports ball" mask from COCO-OneFormer. Using this mask as a prompt for SegGPT resulted in the lowest average quantitative scores (Dice/IoU), suggesting it was a less effective visual cue.



(a) Best Prompt ('1112.jpg')

This prompt resulted in a wider distribution of Dice scores, with a noticeable peak in the 0.6-0.8 range and a higher median (0.253), indicating more frequent successful segmentations.

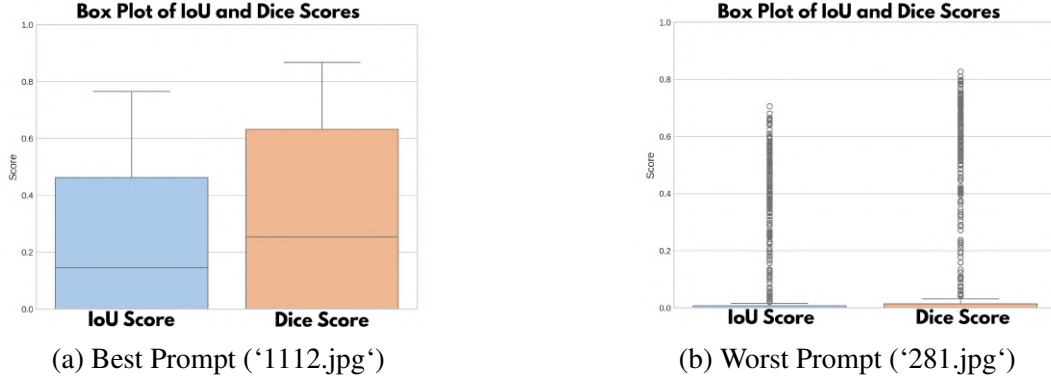


(b) Worst Prompt ('281.jpg')

This prompt led to scores heavily concentrated near zero (median 0.000), suggesting frequent failure to overlap with the ground truth, although some outliers exist.

Figure 4.7: Comparison of Dice Score Distributions for the Best and Worst Performing Prompts.

summaries (median, IQR, outliers) for both IoU and Dice scores between the best and worst prompts.



The box plot for the best prompt shows higher median scores and a wider interquartile range (IQR) compared to the worst case, reflecting more consistent, albeit moderate, performance.

For the worst prompt, the medians are at zero, and the IQR is extremely compressed near the bottom, highlighting the poor central tendency despite the presence of high-scoring outliers.

Figure 4.8: Comparison of IoU and Dice Score Box Plots for the Best and Worst Performing Prompts.

These quantitative results, while modest overall, confirm that: (a) the guided SegGPT approach can produce segmentations with measurable overlap with the ground truth, unlike the baseline OneFormer, and (b) the quality of the segmentation is highly sensitive to the specific visual prompt used, as visually suggested by the differences between the prompts shown in Figures 4.5 and 4.6.

4.2 Discussion of Results

The results from Phase 1, illustrated in Figures 4.1 and 4.2, confirm the difficulty of zero-shot domain transfer for medical image segmentation using models pre-trained solely on natural images. The failure of OneFormer highlights the significant domain gap between natural scenes and medical scans like those in the BraTS-derived dataset [30]. The "sports ball" observation (verified in Figure 4.3), though rare, intriguingly suggests that underlying geometric or textural features learned from natural images can occasionally resonate with visually distinct structures

(like tumors) in the target medical domain, even if the semantic interpretation is completely incorrect.

The Phase 2 results are more nuanced. The consistent qualitative improvement achieved by SegGPT when prompted (Figure 4.4) demonstrates the core principle hypothesized: visual prompting using even flawed, automatically generated masks (like those shown in Figures 4.5 and 4.6) can steer a context-aware model towards segmenting a relevant target region. SegGPT successfully interpreted the visual characteristics presented in the "sports ball" mask to identify and segment similar-looking tumor regions in target images, offering a significant improvement over the scattered and irrelevant baseline outputs.

However, the quantitative results (Figures 4.7 and 4.8) reveal a critical dependency on the specific prompt used. The significant variation in average Dice/IoU scores between the best prompt (derived from Figure 4.5) and the worst prompt (derived from Figure 4.6) indicates that the quality and representativeness of the "weak signal" prompt mask are crucial. A prompt mask that happens to capture the target tumor's shape, boundary characteristics, or internal texture more faithfully leads to better quantitative outcomes. Conversely, a less representative or perhaps noisier prompt mask leads to poorer quantitative overlap, even resulting in near-zero median scores in the worst case observed. This variability underscores a major challenge for practical application: while the principle works, relying on serendipitously "good" misclassifications is not robust.

It is important to reconcile the consistent qualitative improvement with the variable, often low, quantitative scores. The qualitative assessment focused on whether SegGPT produced a more relevant and coherent tumor segmentation than OneFormer's baseline attempt. The quantitative scores, however, measure the pixel-level overlap with the precise ground truth boundaries. Therefore, SegGPT could be consistently better than the baseline qualitatively (producing a plausible tumor-like blob where OneFormer produced nothing relevant), while still achieving only modest overlap with the exact ground truth boundaries quantitatively. The comparison between the best and worst quantitative results highlights that even among the rare "sports ball"

instances, some are inherently better visual cues than others for the SegGPT model, perhaps due to cleaner boundaries or more representative internal patterns in the prompt mask itself.

4.3 Implications

The findings, integrating both qualitative success (Figure 4.4) and quantitative variability (Figures 4.7, 4.8), offer several implications. Firstly, they reinforce the potential of analyzing model failures (like those shown in Figures 4.1-4.3) not just as errors, but as potential sources of useful, albeit weak, signals, particularly in cross-domain scenarios. Secondly, they strongly highlight the power and flexibility of visually prompted models like SegGPT (demonstrated generally in Figure 3.2), but crucially also emphasize their sensitivity to prompt quality – a factor that needs careful consideration in practical applications. Thirdly, while the specific "sports ball" signal is too rare and variable for direct practical application, the experiment demonstrates a mechanism for inference-time, zero-shot adaptation without requiring target domain labels or specific adaptation training. Future work should focus on identifying more common or robust "weak signals" from generalist models and developing methods to automatically select or generate effective prompts. This study serves as a compelling proof-of-concept for leveraging inter-model communication, guided by unusual cross-domain signals, for challenging unsupervised tasks in specialized domains like medical imaging.

Chapter 5

Conclusion

Chapter 5: Conclusion

This chapter concludes the thesis by summarizing the research conducted, reiterating the key findings, acknowledging the study's limitations, and proposing directions for future research based on the outcomes and identified gaps.

5.1 Concluding Remarks

This thesis embarked on an exploration into unsupervised medical image segmentation, specifically targeting a brain tumor dataset derived from the BraTS challenge [30], motivated by the significant challenges associated with acquiring large annotated medical datasets. The core objective was to investigate whether general-purpose vision models, pre-trained on natural images, could yield useful segmentation results in a zero-shot setting, and crucially, whether their failure modes could be leveraged constructively.

The initial phase involved applying three pre-trained OneFormer models [22] directly to unlabeled brain tumor image slices. As anticipated, this zero-shot approach yielded poor results concerning tumor segmentation, confirming the substantial domain gap and the inability of these models to directly recognize relevant clinical structures (illustrated in Figures 4.1 and 4.2). However, detailed manual analysis of the COCO-OneFormer outputs led to a serendipitous discovery: a rare but consistent misclassification where tumor regions were labeled as "sports ball" (verified in Figure 4.3).

This observation formed the basis of the central hypothesis: that this semantically incorrect mask, potentially triggered by shape similarities, contained a "weak signal" that could guide a visually prompted model. Testing this hypothesis, we employed the SegGPT model [39] (whose general mechanism is shown in Figure 3.2), using specific "sports ball" masks as visual prompts. The results demonstrated a clear qualitative improvement (Figure 4.4); SegGPT, guided by these masks, produced tumor segmentations that were subjectively superior in terms of boundary adherence and overall relevance compared to the baseline OneFormer outputs. Quantitative analysis using Dice and IoU scores confirmed that this approach generated segmentations with measurable, albeit modest and variable, overlap with the ground truth (visualized in Figures 4.7

and 4.8). This represents a significant step beyond the effectively zero overlap achieved by the baseline OneFormer in identifying the target tumor region. However, the quantitative results also highlighted a strong sensitivity to the quality of the specific prompt mask used.

In conclusion, this research successfully demonstrated a proof-of-concept: leveraging a specific, domain-shifted misclassification from one pre-trained model (OneFormer) as a visual prompt for another (SegGPT) can lead to markedly improved, unsupervised segmentation results in a specialized domain like medical imaging. While the practical applicability of the specific "sports ball" signal is limited by its rarity and the variability in downstream performance, the study highlights a novel mechanism for inference-time adaptation and inter-model synergy, offering a potential pathway for low-resource segmentation tasks.

5.2 Limitations

Despite the promising findings, this study has several limitations that should be acknowledged:

- **Rarity and Scalability:** The central "sports ball" signal was extremely rare (12 instances in 1500 images processed), making this specific approach impractical for widespread application. Its generalizability beyond these few cases identified in the dataset [30] is unknown.
- **Prompt Variability and Robustness:** The quantitative results showed high sensitivity to the specific prompt mask used (Figures 4.7, 4.8), with performance ranging significantly. This lack of robustness indicates that relying on serendipitous "good" misclassifications is unreliable.
- **Modest Quantitative Performance:** While demonstrating improvement over baseline, the absolute Dice/IoU scores achieved are modest compared to supervised benchmarks for BraTS [30], indicating significant room for improvement.
- **Qualitative Emphasis:** The assessment of improvement relied heavily on qualitative

visual inspection (Figure 4.4), which inherently involves subjectivity.

- **Lack of Comparative Analysis:** The study did not directly compare this novel prompting approach against established unsupervised domain adaptation (UDA) or few-shot segmentation techniques applied to the same dataset.
- **Limited Exploration of Signals:** Only the "sports ball" misclassification from the COCO-OneFormer model was investigated as a prompt source (Figure 4.2). Other misclassifications or outputs from other OneFormer variants (ADE20K, Cityscapes) were not explored as potential prompts.
- **Mechanism Understanding:** The study did not delve into the underlying feature representations to fully understand why the "sports ball" mask occurred or why certain prompts (Figures 4.5, 4.6) performed better than others for SegGPT.
- **Supplementary Visuals:** Additional visual examples supporting the findings (e.g., more SegGPT results, examples for all 12 'sports ball' cases) could be included in an Appendix for completeness but were omitted from the main text for brevity.

5.3 Recommendations for Further Research

Based on the findings and limitations of this thesis, several avenues for future research emerge, aiming to build upon this proof-of-concept and address its current shortcomings.

Firstly, future work should focus on enhancing the core "weak signal" prompting mechanism itself. This involves moving beyond the serendipitous "sports ball" discovery by developing methods to systematically identify other, potentially more common or robust, characteristic misclassifications or feature patterns from generalist models that reliably correlate with target structures in the medical domain. Furthermore, investigating strategies for automatic prompt engineering – selecting the most effective prompt mask from several candidates or even refining noisy masks – would be crucial for improving consistency. Exploring ways to combine

multiple weak signals, perhaps from different layers or even different source models, could also lead to more robust guidance for the target segmentation model.

Secondly, the generalizability and robustness of this approach need thorough investigation. This requires broader quantitative evaluation across a larger set of target images from the dataset [30], employing statistical analysis and potentially additional metrics like the Hausdorff distance to better capture boundary accuracy. Crucially, the methodology should be applied and tested on diverse medical imaging datasets beyond BraTS derivatives, encompassing different anatomical regions (e.g., abdominal organs, lungs), imaging modalities (e.g., CT, Ultrasound), and pathologies. Testing the pipeline with different large pre-trained models as the signal source and alternative prompt-based models for segmentation would also help assess the universality of the observed phenomenon.

Thirdly, a deeper understanding of the underlying mechanisms is warranted. Feature-level analysis should be conducted to investigate why certain cross-domain features (like those triggering the "sports ball" label) are activated by specific anatomical or pathological structures. Understanding why certain prompt masks (Figures 4.5, 4.6) lead to significantly better downstream segmentation performance than others could provide valuable insights for designing more effective prompts, potentially moving from purely empirical observation to a more theoretically grounded approach.

Finally, this novel weak-signal prompting method needs to be contextualized within the existing landscape of segmentation techniques. Direct comparative studies should be performed against established unsupervised domain adaptation (UDA) methods and relevant few-shot learning approaches. Benchmarking the performance, data requirements, and computational costs would clarify the specific advantages, disadvantages, and potential niches where this inference-time adaptation strategy might prove most valuable.

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